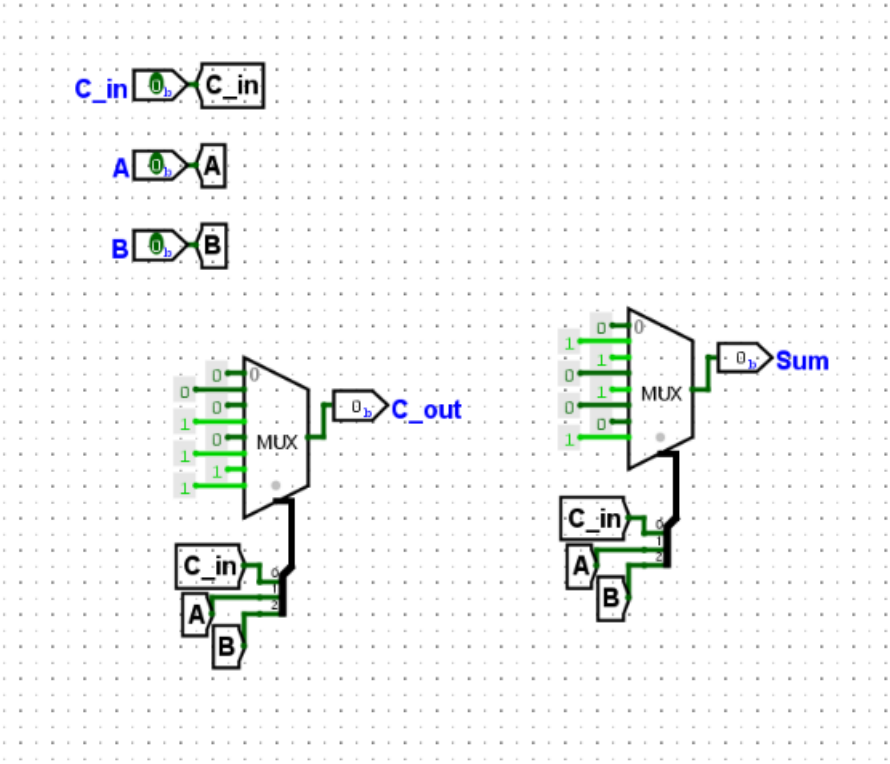
Lab 5

The goal of this lab was to design a full adder using multiplexors and decoders.

A full adder takes 3 bit numbers, adds them, and yields a 2 bit sum. Usually made up of two operands and a carry in bit from previous stages. It produces a two bit output, Carry out and sum. If you have multiple adders, the two bits produced by the adder can be used as the Sum being the bit value for that digit, and the carry out would be the carry in for the next digit.

C_in	A	B	Sum	C_out
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

This is the truth table for a full adder that we used to design our circuits

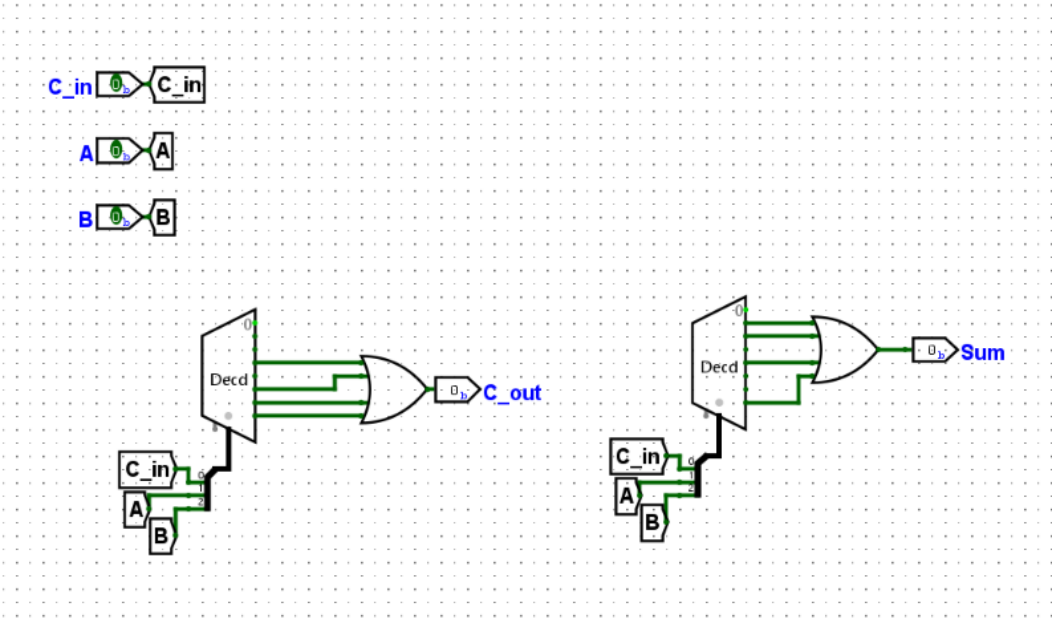


This is a full adder made with 3-1 MUX's.

For the C_out MUX, if any 1 input was high, or if all 3 inputs were high, C_out would be high.

For Sum, if any 2 inputs were high at the same time, or all 3, Sum would be high.

This follows our truth table



This is our full adder made of 3-8 decoders. We selected the pins that would be high in the correct conditions for an adder, and fed them to an or gate, so C_out and Sum would be high at any point on of the correct pins was.